Audi Onboarding Power BI Dashboard Guide

Oct 5, 2026 · @Maryam

What you get

The cleaned file `Audi_Clean.xlsx` has exactly four tables, all to load. Everything the dashboard needs is in those four tables, joined by `EmployeeID`.

Table	Rows	One row per	Use it for
Dim_Employee	39	person	Who, position, wave, start date, PKI / Laptop / Audi ID status, onboarding progress
Dim_Date	92	calendar day	Date slicers, week and month axes
Fact_Attendance	906	employee per recorded day	Attendance %, lateness, leave, the daily attendance grid
Fact_HO_Schedule	560	employee per scheduled day	Site vs Home vs Home Exception, weekly schedule

Sheets dropped from the output, and why:

- **PKI, Laptop Tracker, Audi ID, Training Plan** each hold one value per person, so they became columns of Dim_Employee (`PKI Status`, `Laptop Status`, `Audi User ID`, `Training Task`, …). One table is easier to slice than four.
- **Training Room** is room bookings, not employee tracking.
- **Tools Tracker** holds no data, only a title and a mailing list address.
- **About** was replaced by this guide.

The script prints its fixes and name matches in the terminal instead of saving a log sheet.

Steps 1-3: Prepare and load the data

Step 1: Refresh the clean file. Whenever `Audi.xlsx` is updated, run:

```
python clean_audi_onboarding.py "C:\Onboarding\Audi.xlsx"
"C:\Onboarding\Audi_Clean.xlsx"
```

Always write to the same output path, so Power BI's Refresh button picks up the new data.

Step 2: Load the tables.

1. Open Power BI Desktop, then **Home > Get data > Excel workbook** and pick `Audi_Clean.xlsx`.
2. In the Navigator, tick the four items with the **table icon**: `Dim_Employee`, `Dim_Date`, `Fact_Attendance`, `Fact_HO_Schedule`. Skip the sheet-icon duplicates.
3. Click **Transform Data** (not Load), so you can check types first.

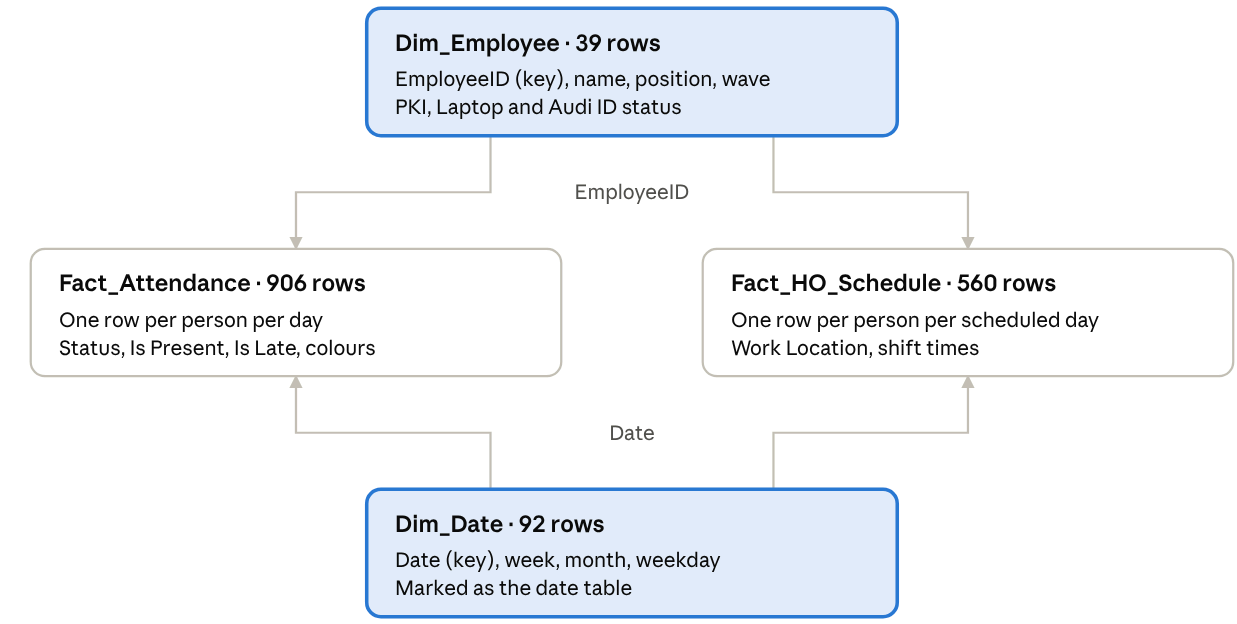
Step 3: Check data types in Power Query. Select each column and set the type from the column header icon:

Table	Column	Type
All	`Date`, `Start Date`, `Week Start`	Date (not Date/Time)
Fact_HO_Schedule	`Shift Start`, `Shift End`	Time
All	every `Is ...` and `... Flag` column, `Status Order`, `Wave`, `Onboarding Steps Done`, `ISO Week`	Whole Number
Fact_HO_Schedule	`Shift Hours`	Decimal Number
Everything else	names, statuses, colours	Text

Then **Home > Close & Apply**.

Step 4: Build the data model

The model is a star: two lookup tables (Dim_Employee, Dim_Date) filter two fact tables. Go to **Model view** (third icon on the left).



data model · 2 lookup tables, 2 fact tables, 4 relationships

Each lookup table (highlighted) filters both fact tables, so one Employee or Date slicer drives every page.

1. Delete any relationships Power BI auto-detected, so you start clean.
2. Drag to create these four relationships. Each is **Many-to-one (*:1)**, cross-filter direction **Single**:
   a. `Dim_Employee[EmployeeID]` to `Fact_Attendance[EmployeeID]`
   b. `Dim_Employee[EmployeeID]` to `Fact_HO_Schedule[EmployeeID]`
   c. `Dim_Date[Date]` to `Fact_Attendance[Date]`
   d. `Dim_Date[Date]` to `Fact_HO_Schedule[Date]`
3. Right-click `Dim_Date` > **Mark as date table** > choose `Date`.
4. Sort text columns by their number columns (select the column, **Column tools > Sort by column**):
   a. `Dim_Date[Month]` by `Month Number`
   b. `Dim_Date[Day Name]` and `Day Short` by `Day Of Week Number`
   c. `Fact_Attendance[Status]` by `Status Order`

5. Hide the fact tables' `EmployeeID` and `Employee Name` columns (right-click > Hide in report view). Always use `Dim_Employee[Employee Name]` in visuals, so one slicer filters every page.
6. Set **Summarization = Don't summarize** on `Wave`, `ISO Week`, `Year` and `Status Order`, so they never show as sums.
7. Create an empty table to hold measures: **Home > Enter data**, name it `_Measures`, click Load. Put every measure from Step 5 there.

Ibrahim Eltohamy appears only in the Audi ID sheet. The measures below count only people with `In Attendance Sheet = "Yes"`, so he doesn't inflate headcount.

Step 5: Add the measures

Select `_Measures`, click **New measure**, paste one line, press Enter, repeat. After the first measure exists, delete the placeholder `Column1` so `_Measures` moves to the top of the Data pane. Format each `%` measure as **Percentage, 0 decimals** (Measure tools > Format).

People and onboarding (not affected by date slicers, by design):

```
Headcount = CALCULATE ( COUNTROWS ( Dim_Employee ), Dim_Employee[In
Attendance Sheet] = "Yes" )
PKI Received = CALCULATE ( [Headcount], Dim_Employee[PKI Status] = "Received"
)
PKI Pending = [Headcount] - [PKI Received]
PKI % = DIVIDE ( [PKI Received], [Headcount] )
Laptops Received = CALCULATE ( [Headcount], Dim_Employee[Laptop Status] =
"Received" )
Laptops Pending = [Headcount] - [Laptops Received]
Laptop % = DIVIDE ( [Laptops Received], [Headcount] )
Audi IDs Issued = CALCULATE ( [Headcount], Dim_Employee[Has Audi ID] = "Yes"
)
Audi IDs Pending = [Headcount] - [Audi IDs Issued]
Audi ID % = DIVIDE ( [Audi IDs Issued], [Headcount] )
Fully Onboarded = CALCULATE ( [Headcount], Dim_Employee[Onboarding Status] =
"Complete" )
Fully Onboarded % = DIVIDE ( [Fully Onboarded], [Headcount] )
```

Attendance:

```
Working Days = SUM ( Fact_Attendance[Is Working Day] )
Present Days = SUM ( Fact_Attendance[Is Present] )
Attendance % = DIVIDE ( [Present Days], [Working Days] )
Late Days = SUM ( Fact_Attendance[Is Late] )
Late % = DIVIDE ( [Late Days], [Present Days] )
Tech Issue Days = SUM ( Fact_Attendance[Is Tech Issue] )
Planned Leave Days = CALCULATE ( COUNTROWS ( Fact_Attendance ),
Fact_Attendance[Status Category] = "Planned Leave" )
Unplanned Absences = CALCULATE ( COUNTROWS ( Fact_Attendance ),
Fact_Attendance[Status Category] = "Unplanned Absence" )
First Days = SUM ( Fact_Attendance[Is First Day] )
Status Code = SWITCH ( SELECTEDVALUE ( Fact_Attendance[Status] ), "Attended",
"P", "Attended Late", "L", "Off", "-", "Annual Leave", "AL", "Sick Leave",
"S", "Emergency Leave", "E", "No Call No Show", "NCNS", "Home Office Issue",
"HO" )
Status Colour = SELECTEDVALUE ( Fact_Attendance[Status Color] )
```

H-O Schedule:

```
Scheduled Days = CALCULATE ( COUNTROWS ( Fact_HO_Schedule ),
Fact_HO_Schedule[Schedule Type] = "Shift" )
On-site Days = SUM ( Fact_HO_Schedule[Is On Site] )
Home Days = CALCULATE ( COUNTROWS ( Fact_HO_Schedule ), Fact_HO_Schedule[Work
Location] = "Home" )
Home Exception Days = CALCULATE ( COUNTROWS ( Fact_HO_Schedule ),
Fact_HO_Schedule[Work Location] = "Home Exception" )
On-site % = DIVIDE ( [On-site Days], [Scheduled Days] )
Location Code = SWITCH ( SELECTEDVALUE ( Fact_HO_Schedule[Work Location] ),
"Site", "S", "Home", "H", "Home Exception", "HE", SELECTEDVALUE (
Fact_HO_Schedule[Schedule Type] ) )
Location Colour = SELECTEDVALUE ( Fact_HO_Schedule[Location Color] )
```

Quick check with the current file: Attendance % should show **99%**, Late % **11%**, PKI Received **20 of 38**, Laptops Received **37 of 38**.

Step 6: Build the five pages

Every page shares one slicer bar across the top: **Date** (`Dim_Date[Date]`, style Between), **Wave**, **Position** and **Employee Name** (all from Dim_Employee, style Dropdown). Build it once on page 1, then copy it to the other pages and choose **Sync** when Power BI asks.

Page 1: Overview

Answers "how is onboarding going overall?" in 10 seconds.

1. **Six Card visuals** in one row: `Headcount`, `Attendance %`, `Late %`, `PKI %`, `Laptop %`, `Fully Onboarded %`.
2. **Clustered bar chart** "Onboarding progress": add `PKI Received`, `Laptops Received`, `Audi IDs Issued` and `Fully Onboarded` as values, no axis field. Turn on data labels.
3. **Line chart** "Attendance trend": X-axis `Dim_Date[Date]`, Y-axis `Attendance %`; add `Late %` as a second line.
4. **Stacked column chart** "New joiners": X-axis `Dim_Employee[Start Month]`, Y-axis `Headcount`, Legend `Wave`.
5. **Donut chart** "Onboarding status": Legend `Dim_Employee[Onboarding Status]`, Values `Headcount`.

Page 2: Attendance

Replaces the colour grid in the Attendance sheet.

1. **Matrix** "Daily attendance": Rows `Dim_Employee[Employee Name]`, Columns `Dim_Date[Date]`, Values `Status Code`. In Format > Cell elements, turn on **Background color**, Format style **Field value**, field `Status Colour`. Turn off row and column subtotals. The result looks like the original sheet, with one cell per person per day.
2. **Stacked column chart** "Status by week": X-axis `Dim_Date[Week Label]`, Y-axis count of `Fact_Attendance[Status]`, Legend `Fact_Attendance[Status]`. Filter out `Off` in the visual's filter pane.
3. **Table** "Attendance by person": `Employee Name`, `Working Days`, `Present Days`, `Attendance %`, `Late Days`, `Tech Issue Days`, `Unplanned Absences`. Sort by `Late Days` descending. Add data bars to `Late Days`.
4. **Cards**: `Late Days`, `Tech Issue Days`, `Planned Leave Days`, `Unplanned Absences`.

Page 3: Onboarding (PKI, Laptop, Audi ID)

Answers "who still needs what?".

1. **Three Gauge visuals**: `PKI Received`, `Laptops Received` and `Audi IDs Issued`, each with Maximum value `Headcount`.

2. **Table** "Onboarding checklist": `Employee Name`, `Position`, `Start Date`, `Wave`, `PKI Status`, `Laptop Status`, `Has Audi ID`, `Training Mode`, `Onboarding Steps Done`. Apply conditional **Background color > Rules** on the status columns: `Received` / `Yes` green, `Not Received` / `No` red.
3. **Slicer** on `Onboarding Status`, so you can show only `In Progress` and `Not Started`: your to-do list for the week.
4. **Clustered column chart** "Pending by item": values `PKI Pending`, `Laptops Pending`, `Audi IDs Pending`.

Page 4: H-O Schedule

Replaces the weekly Site / Home colour grid.

1. **Matrix** "Weekly schedule": Rows `Employee Name`, Columns `Dim_Date[Date]`, Values `Location Code`, background colour by field value `Location Colour` (green = Site, white = Home, yellow = Home Exception, orange = leave).
2. **Stacked bar chart** "Location mix by week": Y-axis `Fact_HO_Schedule[Week Number]`, X-axis `Scheduled Days`, Legend `Work Location`.
3. **Cards**: `On-site %`, `On-site Days`, `Home Days`, `Home Exception Days`.
4. **Column chart** "On-site by weekday": X-axis `Dim_Date[Day Name]`, Y-axis `On-site Days`. This shows which days the office is busiest.

Page 5: Employee detail (drill-through)

One person's full picture, opened by right-clicking their name on any page.

1. Add `Dim_Employee[Employee Name]` to this page's **Drill through** well.
2. **Multi-row card**: `Employee Name`, `Position`, `Start Date`, `Wave`, `Email`, `Audi User ID`, `Training Task`, `Training Owner`.
3. **Cards**: `Attendance %`, `Late Days`, `On-site Days`.
4. **Table**: `Fact_Attendance[Date]`, `Fact_Attendance[Status]`, `Late Type`, `Color Note`, sorted by date. This is the person's day-by-day attendance history.
5. **Table**: `PKI Status`, `Laptop Status`, `Has Audi ID`, `Onboarding Status`.

Step 7: Polish, refresh and check

Colours. The cleaned file copies each cell's colour from the original sheets, so the team recognises them. The coloured columns in Excel are painted the same way. The cleaned file stores them as hex codes in `Status Color` and `Location Color`:

Meaning (source legend)	Colour	Hex
Attended / Site	green	#92D050
Attended, first day	pink-purple	#D86ECC
Attended late	light green	#D9F2D0
Late: 30 min site / 15 min HO	coral	#F5496F
Home exception (attendance)	light peach	#FAE3D6
Annual leave	amber	#FFC000
Sick leave / Home exception (schedule)	yellow	#FFFF00
Emergency leave	blue	#00B0F0
No call no show	red	#FF0000
Off	grey	#AEAEAE
Home office issue	light grey	#D1D1D1
Home (schedule)	white	#FFFFFF

In every chart that uses `Status` or `Work Location` as its legend, set each series to these colours (Format > Columns > Colors). For everything else, apply the theme file `Audi_Onboarding_Theme.json`: **View > Themes > Browse for themes**, then pick the file. It sets the grey page background, white rounded visual cards, navy table headers, card and slicer fonts, and turns off matrix subtotals.

Layout tips.

- Canvas 16:9. Slicer bar across the top, cards in the second row, charts below.
- Give each page a title text box that states what it answers, for example "Who still needs PKI, laptop or Audi ID?".
- On the two grid matrices, set column width to about 40 px and turn **Word wrap off**, so a month fits on screen.

- Add **Last refresh** to Page 1: a card on a measure `Data As Of = CALCULATE ( MAX ( Fact_Attendance[Date] ), Fact_Attendance[Is Working Day] = 1 )`.

Weekly refresh routine.

1. Update `Audi.xlsx` as usual.
2. Run the script (Step 1), writing to the same `Audi_Clean.xlsx`.
3. Read the notes the script prints in the terminal, especially **"Fuzzy match - please verify"** and **"Added as new employee"**. Fix a wrong match by adding the name to `NAME_ALIASES` at the top of the script, then rerun.
4. In Power BI, click **Home > Refresh**, then save.

Checks after the first build.

- [ ] Headcount card shows 38
- [ ] Attendance % shows 99% and Late % 11% with no slicers applied
- [ ] Daily attendance matrix matches the Attendance sheet for two people you know
- [ ] Weekly schedule matrix shows week 41 (5-9 Oct) mostly green Mon-Wed, white Thu-Fri
- [ ] Right-clicking a name > Drill through > Employee detail opens the person's page
- [ ] Selecting an employee in the slicer filters all five pages